AI & GENERATIVE AI TRACK / PROJECT 04 OF 05

# LearnMate

PHASE 03 OF 08 | PROJECT DESIGN PHASE

## Project overview

LearnMate helps students work with scattered course materials through a REST API for uploads and AI-generated learning resources. The brief describes secure accounts, MongoDB storage, Gemini-powered content generation, and a dashboard-facing backend for saved study resources.

## Project description

A study-support API for uploading learning materials and generating summaries, flashcards, quizzes, and personalized study plans.

## Project status

Not stated in the source brief. Update this document when the team records an agreed status.

## Project capabilities

- Study material uploads
- AI summaries and flashcards
- Practice quizzes
- Personalized study plans

## Phase description

Check problem-solution fit and describe the proposed solution and its architecture.

## Project-specific phase focus

Design the journey from uploaded content to reviewable, reusable learning resources.

## Phase status and progress

Status: Not stated in the source material.
Progress: Not recorded in the source material.

## Tasks and requirements

[ ] Complete the problem-solution fit canvas
[ ] Describe the proposed solution and user value
[ ] Diagram the solution architecture

## Relevant documents and data

Project brief: LearnMate - AI assistant for smarter learning and study support..docx

Phase templates:
- Problem-Solution Fit (3. Project Design Phase/Problem-Solution Fit.pdf)
- Proposed Solution (3. Project Design Phase/Proposed Solution.pdf)
- Solution Architecture (3. Project Design Phase/Solution Architecture.pdf)

## Project technology stack

- Node.js
- Express.js
- MongoDB
- Mongoose
- Gemini AI

## Phase notes and decisions